Employee Laptop Request Management System

Requirement Analysis

Project Name	Employee Laptop Request Management System
Project Type	Individual Project
Document Date	12 September 2026
Platform	ServiceNow

1. Purpose and Scope

This document defines the functional and non-functional requirements for the Employee Laptop Request Management System implemented on ServiceNow. The scope covers request submission, information capture, classification, annual duplicate prevention, manager approval, status handling, email notifications, role-based access and validation.

2. Stakeholders and User Roles

Stakeholder / role	Responsibilities
Employee / requester	Submit a laptop request, provide accurate details and view request status/outcome.
IT Manager	Review assigned requests and approve or reject them through the approval process.
IT Employee	Create, read and write permitted laptop request records as configured by role and ACL.
ServiceNow administrator	Maintain application configuration, roles, ACLs, Flow Designer and update sets.

3. Functional Requirements

ID	Requirement	Description
FR-01	Laptop request submission	Provide a centralized Record Producer through the Service Portal.
FR-02	Information capture	Capture requester, employee email, request date, short description, manager, request type and reason.
FR-03	Classification	Support Basic / Advance and Software Development / Personal Use / Other reason values.
FR-04	Duplicate prevention	Block a second request by the same requester within the current calendar year.
FR-05	Manager approval	Route a valid request to the referenced manager for approval or rejection.
FR-06	Status management	Maintain Pending Approval, Approved and Rejected states.
FR-07	Notifications	Notify the manager of a pending decision and the requester of the outcome.
FR-08	Role-based access	Apply it_employee and it_manager permissions through ACLs.
FR-09	Approval relationship	Associate the request with the corresponding sysapproval_approver record.
FR-10	Validation and deployment	Validate form, duplicate policy, flow, notifications, access and update-set capture.

4. Non-Functional Requirements

ID	Quality attribute	Requirement
NFR-01	Usability	Provide a clear form with understandable labels and minimal unnecessary steps.
NFR-02	Security	Use ServiceNow roles and ACLs to restrict access by responsibility.
NFR-03	Reliability	Maintain consistent request, approval and status information.
NFR-04	Performance	Provide responsive form and record operations under normal platform use; no benchmark is claimed.
NFR-05	Availability	Rely on the availability of the configured ServiceNow instance.
NFR-06	Scalability	Support growth in employees and requests without changing the core lifecycle design.

5. User Stories and Acceptance Criteria

ID	User story	Acceptance criteria
US-01	As an employee, I can submit a laptop request.	A valid form creates one request record.
US-02	As an employee, I can provide request type and reason.	Selected choice values are stored correctly.
US-03	As a requester, I can avoid duplicate submissions.	A second request in the same calendar year is blocked.
US-04	As a manager, I receive approval requests.	A valid submission creates/routs an approval to the referenced manager.
US-05	As a manager, I can approve or reject.	The approval decision is recorded and processed.
US-06	As an employee, I receive the final outcome.	An outcome notification is generated after the decision.
US-07	As an IT employee, I can manage permitted records.	Create/read/write access follows the configured ACLs.
US-08	As an IT manager, I can view requests.	Read access is allowed while create/write is restricted by the design.
US-09	As an administrator, I can trace approval records.	The request and approval record relationship can be inspected.
US-10	As an administrator, I can validate deployment readiness.	Required components are captured and checked in an update set.

6. Data Flow Diagrams

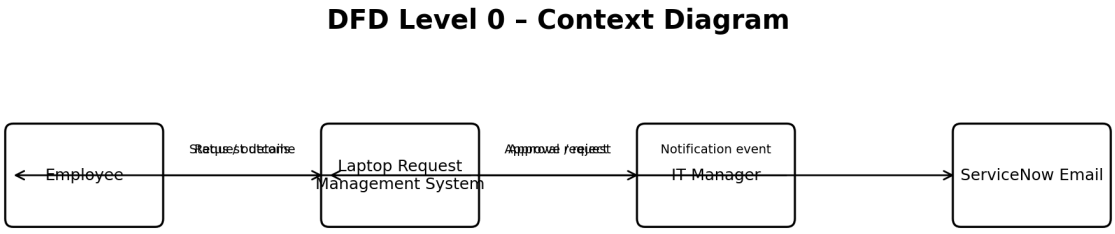


Figure 1. Level 0 context diagram.

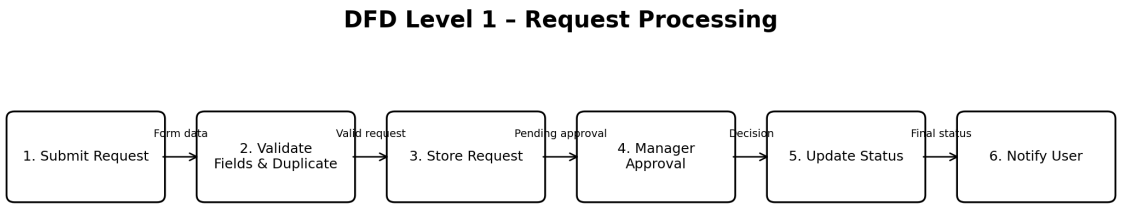


Figure 2. Level 1 request-processing data flow.

7. Customer Journey Map

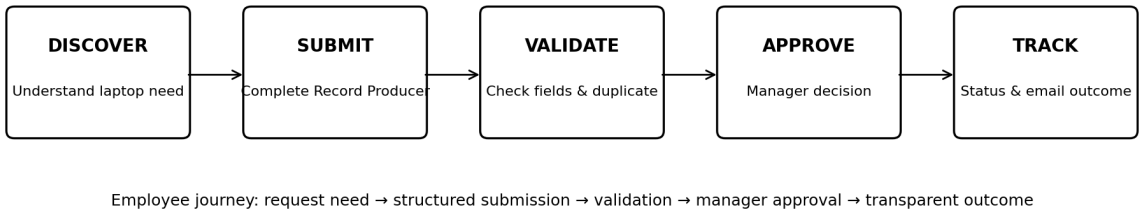


Figure 3. Employee request journey from discovery to reuse.

8. Technology Stack

Layer	Technology / component	Purpose
Platform	ServiceNow scoped application	Application runtime and service-management platform.
User interface	Service Portal Record Producer	Employee request entry point.
Data	Custom Laptop Request Form table	Stores request information.
Business logic	Business Rule / GlideRecord	Checks annual duplicate requests before insert.
Workflow	Flow Designer / Ask for Approval	Routes manager decision and updates status.
Approval	sysapproval_approver	Stores approval activity.
Security	it_employee, it_manager, ACLs	Controls record operations.
Communication	ServiceNow notifications	Sends approval and outcome messages.
Deployment	Update Sets / XML	Captures configuration for transfer.

Note: Requirement note: the annual duplicate policy is defined as one request per requester per current calendar year. Confirm the date field and requester identity used in the live Business Rule before acceptance.